

Here are **detailed Week 7 notes** for **Session X: Model Understanding** plus **End-to-End ML Case Study Part 1**.

Week 7 — Model Understanding + End-to-End ML Part 1

Machine Learning and Advanced Analytics

1. Main aim of model understanding

Model understanding is about using a trained predictive model to understand:

What, if observed, helps predict the output variable or class?

This is still **observational machine learning**, not causal inference. We are not saying “this variable causes the outcome.” We are saying “this variable helps the model predict the outcome.”

For example, in churn prediction, variable importance can help us understand what kinds of customer behaviour are associated with churn. This can support targeting, communication style, campaign design, and customer “pen portraits.” The Session 10 slides frame this as moving from **feature identification** to **customer/problem understanding**.

Part A — Variable Importance for Model / Problem Understanding

2. Why use variable importance for understanding?

Variable importance helps us identify which features matter most to a trained model.

It can be used for:

Purpose	Meaning
Model refinement	Remove weak or unnecessary features
Problem understanding	Understand which observed behaviours predict the outcome
Customer profiling	Build pen portraits of high-risk or high-value groups
Intervention thinking	Identify candidate variables to investigate further
Cost reduction	Avoid collecting expensive but low-value variables

In Session 10, the churn case study explains that variable importance can be used to understand churners versus non-churners, support communication tone/campaign style, and reason about possible interventions even before a causal effect is measured.

3. Variable importance is not causal importance

A very important exam point:

A feature can be important for prediction without being the cause of the outcome.

Example:

Umbrella ownership may predict rain exposure,
but umbrellas do not cause rain.

In business analytics, this distinction matters because predictive models answer:

If I observe X, can I predict Y?

Causal models answer:

If I change X, will Y change?

For model understanding, we are mostly asking the first question.

4. Why observational understanding is still useful

Even if a feature is not causal or directly changeable, it can still be useful.

For example, we may not be able to change a customer's personality, introversion, impulsiveness, or long-term behaviour type. But if those features help predict product interest or churn risk, we can still use them for:

targeting
segmentation
communication tone
campaign design
advertising strategy
product development

The slides explicitly note that not all factors can be influenced, but knowing that they predict behaviour can still help tailor communications and build pen portraits.

Part B — Summary Statistics vs Variable Importance

5. Summary statistics approach

A simple way to compare two groups is to calculate summary statistics.

Example:

```
Average spend of churners vs non-churners  
Average visits of churners vs non-churners  
Average discount use of churners vs non-churners
```

This is easy and intuitive.

Advantages

```
Simple  
Fast  
Easy to explain  
Useful for initial exploration
```

Problems

Summary statistics treat each feature independently. They do not tell us whether the feature actually helps a predictive model.

A variable may look different between two groups but not matter once other variables are considered.

The slides state that summary statistics can lead to pen portraits based on features that do not matter for most people, because aggregates can hide truths and features are considered independently.

6. Variable importance approach

Variable importance starts from a trained predictive model.

The process is:

1. Train a good predictive model.
2. Calculate feature importance.
3. Identify variables that strongly affect predictions.
4. Analyse those important variables further.
5. Use them to build meaningful group summaries / pen portraits.

This is better than simple summaries because it focuses attention on features that actually contribute to prediction.

7. Simple comparison

Approach	What it tells us	Weakness
Summary statistics	How groups differ on average	May highlight irrelevant differences
Variable importance	Which variables help prediction	Depends on model quality
PDPs	Direction and shape of modelled effect	Shows model behaviour, not causal truth

Part C — Pen Portraits Using Variable Importance

8. What is a pen portrait?

A **pen portrait** is a concise profile of a typical entity, such as a customer, student, store, or product group.

Example:

High churn-risk customers tend to have low recent visit frequency, declining spend, long time since last purchase, and reduced category variety.

In model understanding, pen portraits should not be based only on raw group averages. They should be based on important predictive variables.

9. How to create a model-based pen portrait

A strong method is:

1. Train a predictive model.
2. Check validation/test performance.
3. Calculate variable importance.
4. Select the most important features.
5. Compare these features across predicted/actual groups.
6. Use PDPs to understand direction of effect.
7. Write the customer profile carefully.

Example:

The model suggests churn risk is mainly associated with recency, declining spend, and reduced visit frequency. High-risk customers are not simply low-spending customers; rather, they show a downward pattern compared with their earlier behaviour.

This is stronger than saying:

Churners spend less on average.

because it uses the model to identify which variables matter most.

Part D — Partial Dependence Plots

10. What is a Partial Dependence Plot?

A **Partial Dependence Plot**, or **PDP**, shows how the model's predicted output changes as one feature changes, while averaging over the other features.

In simple terms:

A PDP asks: according to the model, what happens to the prediction when this feature increases?

The slides introduce PDPs using a prediction function such as:

```
purchase_prob = f(price, quality)
```

To understand the effect of `price`, we vary `price` while keeping the other values from the dataset, then average the predictions.

11. Intuition behind PDPs

Suppose the model predicts wine purchase probability using:

```
price  
quality  
brand  
customer type
```

To understand `price`, we create different versions of the data:

```
price = £10  
price = £20
```

```
price = £30  
price = £40
```

For each price value, we ask the model to predict outcomes for many rows. Then we average the predictions.

The result shows whether the model thinks increasing price increases or decreases purchase probability.

12. How to interpret a PDP

A PDP usually has:

```
x-axis: feature value  
y-axis: average model prediction
```

Example interpretation:

```
As price increases, predicted purchase probability decreases.
```

This means the model has learned a negative relationship between price and purchase probability.

The slides explain that the mean line shows how the output changes in response to the input **on average**, while grey ICE-style lines can show how variable this relationship is across observations.

13. PDP example: price and purchase probability

If the PDP slopes downward:

```
Higher price → lower predicted probability of purchase
```

If the PDP slopes upward:

```
Higher value of the feature → higher predicted probability
```

If the PDP is flat:

```
The feature has little average effect on the prediction
```

If the PDP is curved:

The model has learned a non-linear relationship

For example, a PDP may show that price has little effect up to £30, then sharply reduces purchase probability after £40.

14. Important PDP caveat

A PDP shows what the **model thinks**, not necessarily what is true in the real world.

This is a major exam point.

If the model is wrong, the PDP interpretation is also wrong.

The slides explicitly warn that PDPs show the model's view of the world, not the "truth" in the world.

15. PDPs and interactions

PDPs average over other features. This can hide interactions.

Example:

Discount may increase purchase probability for loyal customers,
but have little effect for new customers.

A one-feature PDP may average these two patterns together and produce a misleading middle result.

This is why ICE plots or 2D PDPs can be useful.

16. ICE plots

An **Individual Conditional Expectation plot** shows one line per observation.

A PDP is the average of these individual lines.

ICE plot = many individual response curves
PDP = average response curve

ICE plots are useful because they show whether the feature has a similar effect for everyone or whether the effect varies across observations.

If all ICE lines have similar shape, the PDP is more trustworthy.

If ICE lines are very different, the PDP average may hide important subgroups.

17. 2D PDPs

A 2D PDP shows how predictions change when two features vary together.

Example:

```
Prediction = f(price, quality)
```

A 2D PDP can show interaction effects such as:

```
Price only reduces purchase probability when quality is low.
```

The Session 10 slides include a 2D PDP example about predicting whether a football team member will win Man of the Match, where the interpretation depends jointly on goals scored and distance run.

Part E — Creating PDPs in Python

18. Basic sklearn PDP code

A common way to create PDPs in Python is with scikit-learn:

```
from sklearn.inspection import PartialDependenceDisplay
import matplotlib.pyplot as plt

features = ["price"]

PartialDependenceDisplay.from_estimator(
    estimator=model,
    X=X_valid,
    features=features
)

plt.show()
```

For multiple features:

```
features = ["price", "quality", "customer_age"]

PartialDependenceDisplay.from_estimator(
    estimator=model,
    X=X_valid,
    features=features
)
```



```
plt.show()
```

For a 2D PDP:

```
features = [("price", "quality")]

PartialDependenceDisplay.from_estimator(
    estimator=model,
    X=X_valid,
    features=features
)

plt.show()
```

19. Practical interpretation template

Use this structure when writing about a PDP:

The PDP for [feature] shows that as [feature] increases, the model's predicted probability of [outcome] generally [increases/decreases/remains stable]. This suggests that the model associates [feature] with [higher/lower] risk. However, this should be interpreted as model behaviour rather than a causal effect.

Example:

The PDP for recency shows that as days since last purchase increases, the model's predicted probability of churn rises sharply. This suggests that the model treats recent inactivity as an important churn signal. However, this is an observational model interpretation and should not be read as proof that recency causes churn.

Part F — SHAP Values

20. SHAP values overview

The Session 10 slides also introduce SHAP values as another model understanding method.

SHAP values explain individual predictions.

They ask:

For this specific data point, how much did each feature push the prediction up or down?

So PDPs are more global, while SHAP is often local.

Method	Main use
Variable importance	Which features matter overall?
PDP	How does prediction change as a feature changes?
ICE	Does the feature effect differ by observation?
SHAP	Why did this specific prediction happen?

The slides describe SHAP as local interpretation that explains how the model predicts each point, while also warning that this explains the model, not necessarily the real-world phenomenon.

Part G — Limits of Model Understanding

21. Model understanding vs phenomenon understanding

A key advanced point from Session 10 is that explaining one model is not always the same as understanding the true phenomenon.

Different models can achieve similar performance using different variables.

This means feature importance may depend on:

- algorithm choice
- random seed
- hyperparameters
- correlated variables
- shared predictive information

The slides show that if several models perform equally well, their variable importance scores can differ, meaning phenomenon interpretation may depend on arbitrary modelling choices.

22. Rashomon Set

The **Rashomon Set** is the set of models that perform similarly well.

The idea is:

There may not be one single best explanation.
There may be many equally good predictive relationships.

If multiple models achieve similar accuracy but rely on different variables, then relying on just one model's explanation may be misleading.

The slides define the Rashomon Set as the set of equally performing models within a tolerance of a reference model.

23. Model Class Reliance

Model Class Reliance, or **MCR**, tries to understand variable importance across a class of good models, rather than just one fitted model.

Instead of asking:

How important is this feature in my one model?

MCR asks:

Across many similarly good models, how important could this feature be?

The slides explain MCR as returning lower and upper bounds of variable importance across the Rashomon Set.

This is useful because it gives a more honest view of uncertainty in model explanations.

Part H — End-to-End ML Case Study Part 1

24. What is an end-to-end ML task?

An end-to-end ML task covers the full process from business problem to evaluation and interpretation.

Typical stages:

1. Define the business problem
2. Define the prediction task
3. Identify the unit of prediction
4. Define the target variable
5. Create features
6. Split data correctly
7. Train baseline models
8. Train candidate ML models
9. Tune and evaluate models

10. Interpret the model
11. Justify decisions
12. Communicate results

For temporal prediction, this also includes choosing reference dates and ensuring no leakage.

25. Defining the prediction task

A good ML project starts with a clear prediction task.

Weak version:

Predict demand.

Better version:

Predict store-level product demand for the next 7 days using only information available up to the reference date.

A complete prediction task should define:

entity: customer / product / store / household
reference date: when prediction is made
prediction horizon: how far into the future
target: what is predicted
features: what information is allowed
evaluation metric: how success is measured

26. Example temporal prediction setup

Example:

Business problem:
Improve stock availability.

Prediction task:
Predict next-week demand for each product-store pair.

Entity:
Product-store pair.

Reference date:
End of each week.

Features:

Historical sales, promotions, seasonality, stock levels, price.

Target:
Units sold in the next week.

Evaluation:
WAPE, MAE, RMSE, or service-level impact.

27. Justifying decisions in end-to-end ML

A key expected outcome is understanding **when and how to justify decisions**.

You should justify:

Decision	Example justification
Target definition	Matches the business decision window
Prediction horizon	Reflects when action must be taken
Features	Available before prediction date
Train/test split	Avoids temporal leakage
Metric	Matches business cost
Baseline	Represents simple operational alternative
Model choice	Balances performance and interpretability
Interpretation method	Supports stakeholder understanding

28. Temporal evaluation metrics for demand prediction

Demand prediction often uses regression-style metrics.

MAE

Mean Absolute Error

Average absolute difference between actual and predicted demand.

Good because it is easy to understand.

RMSE

Root Mean Squared Error

Penalises large errors more strongly than MAE.

Good when large errors are especially bad.

MAPE

Mean Absolute Percentage Error

Measures percentage error.

Problem: unstable when actual demand is zero or very small.

WAPE

Weighted Absolute Percentage Error

Usually calculated as:

$$\text{sum absolute errors} / \text{sum actual demand}$$

Often better than MAPE for demand forecasting because it handles aggregation more sensibly.

WMAPE

Weighted version of MAPE, where important products, stores, or periods can receive more weight.

29. MAPE vs WAPE vs WMAPE

Metric	Meaning	Main issue
MAPE	Average percentage error per row	Bad with zero/small actuals
WAPE	Total absolute error divided by total actual demand	Can hide item-level errors
WMAPE	Weighted percentage error	Depends on weight choice

For retail demand prediction, **WMAPE** is often preferred because it is more stable when some products have low or zero demand.

30. Strong exam-style answers

Q1. Why use variable importance instead of summary statistics?

Summary statistics show how groups differ on average, but they do not tell us whether those differences matter for prediction. Variable importance uses a trained predictive model to identify

which features actually contribute to predicting the outcome. This helps create more meaningful customer profiles and avoids focusing on variables that differ between groups but are not useful predictors.

Q2. What does a PDP show?

A Partial Dependence Plot shows how a model's average prediction changes as one feature is varied. It helps identify the direction and shape of the modelled relationship, such as whether predicted churn risk increases as days since last purchase increases. A PDP explains the model's behaviour, not necessarily a causal relationship in the real world.

Q3. What is the difference between PDP and ICE?

A PDP shows the average effect of a feature across all observations. An ICE plot shows the effect separately for each observation. ICE plots reveal whether the feature effect is consistent across individuals or whether the PDP average is hiding different subgroup behaviours.

Q4. How can variable importance support pen portraits?

Variable importance identifies the features that most strongly influence model predictions. These important features can then be compared across groups, such as churners and non-churners, to create a pen portrait. This produces a more model-relevant profile than using all available variables or simple group averages.

Q5. Why must PDPs be interpreted carefully?

PDPs show how the trained model responds to changes in a feature. If the model is inaccurate, biased, or missing important variables, the PDP may also be misleading. PDPs are also not causal; they show modelled associations, not what would happen if the business intervened and changed the feature.

Q6. What is an end-to-end ML case study?

An end-to-end ML case study covers the full machine learning workflow: defining the business problem, creating the prediction task, preparing data, building features, choosing evaluation design, training baselines and models, evaluating performance, interpreting results, and justifying decisions. For temporal prediction, it must also ensure that training uses only information available before the prediction date.

31. One-page revision summary

Model understanding = using trained models to understand what observed features predict the output.

Summary statistics = simple group differences.

Problem: may focus on irrelevant features and ignores interactions.

Variable importance = identifies features that matter to prediction.

Useful for model refinement, customer understanding, and pen portraits.

PDP = shows how average model prediction changes as one feature changes.

ICE = individual version of PDP.

PDP caveat = explains model behaviour, not causal truth.

SHAP = explains individual predictions by feature contributions.

Rashomon Set = many similarly good models may rely on different variables.

MCR = studies variable importance bounds across similarly good models.

End-to-end ML = full workflow from business problem to prediction task,
features,
evaluation, modelling, interpretation, and justification.

Temporal end-to-end ML must avoid leakage by using past data to predict future outcomes.